

HAT-P-32b

R-0197

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_241115 · created 2026-07-16T21:42:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460629.6786 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.012
Transit depth (Rp/Rs)^2	2.64 +/- 0.38 [%]
Orbital Inclination (inc)	89.86 +/- 1.22
Airmass coefficient 1 (a1)	1.009 +/- 0.015
Airmass coefficient 2 (a2)	-0.012 +/- 0.054
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#3 - [481, 363]
Optimal Method	PSF photometry
Transit Duration (day)	0.1008 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.162 $\pm$ 0.012 vs 0.1489 (deviation 1.04 σ)
Duration fit vs published	0.1008 d vs 0.1304 d (deviation 14.82 σ)
Mid-time O-C	11.6 min
Depth SNR	12.8
Residual scatter	1.47 %

Light curve

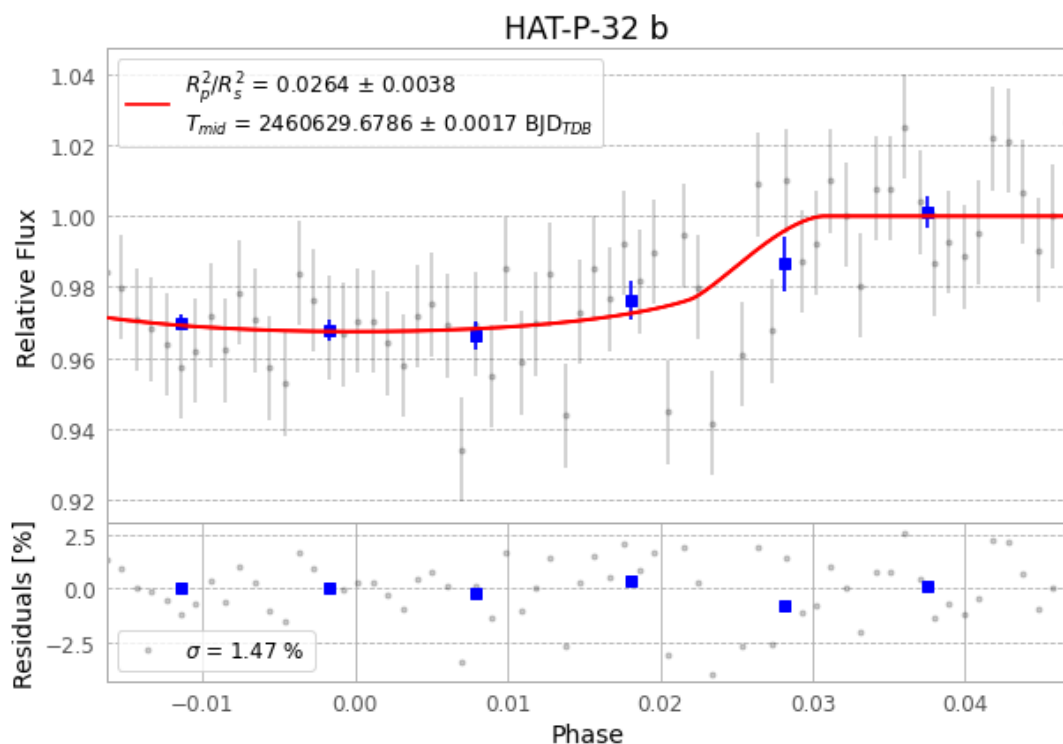


Figure 1 — FinalLightCurve_HAT-P-32 b_2024-11-14.png (SHA-256 dba45b2d9b3942dd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 228], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c0c486d0573c683d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

66 FITS frames (43 MB), HATP-32241115031816.FITS → HATP-32241115063315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-241115.log (e4916c238156...), FinalLightCurve_HAT-P-32 b_2024-11-14.png (dba45b2d9b39...), FinalLightCurve_HAT-P-32 b_2024-11-14.csv (e32660139112...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 21:42Z.